Pre-Calculus Concepts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Limits: an introductory numerical approach

Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression algebraically.

Factor: $\frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$. As $x \to 2$, this approaches $2 + 2 = 4$.

2 [Foundational] • Target: Average rate of change

Find the average rate of change of $f(x) = x^2 + 1$ over the interval $[1, 4]$.

Average rate of change $= \frac{f(4) - f(1)}{4 - 1} = \frac{17 - 2}{3} = \frac{15}{3} = 5$.

3 [Foundational] • Target: Conic sections: ellipses

Find the equation of an ellipse centered at the origin with $a = 5$ (horizontal semi-axis) and $b = 3$ (vertical semi-axis).

$\frac{x^2}{25} + \frac{y^2}{9} = 1$.

4 [Foundational] • Target: Conic sections: ellipses

For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the coordinates of the vertices along the major axis.

Since $16 > 9$, the major axis is horizontal, with $a = \sqrt{16} = 4$. Vertices: $(4, 0)$ and $(-4, 0)$.

5 [Foundational] • Target: Conic sections: hyperbolas

Find the equation of a hyperbola centered at the origin with vertices at $(\pm 4, 0)$ and $b = 3$.

Since the vertices are horizontal, the equation has the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with $a = 4$: $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

6 [Foundational] • Target: Parametric equations

Eliminate the parameter from $x = t + 1$ and $y = t^2$, expressing y in terms of x.

From the first equation, $t = x - 1$. Substitute: $y = (x - 1)^2$.

7 [Foundational] • Target: Sequences and series: sigma notation

Write $4 + 8 + 12 + 16 + 20$ using sigma notation, and evaluate it.

This is $\sum_{k=1}^{5} 4k$. Evaluating: $4(1 + 2 + 3 + 4 + 5) = 4(15) = 60$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Binomial theorem

Use the binomial theorem to expand $(x+2)^4$.

$(x+2)^4 = x^4 + 4x^3(2) + 6x^2(4) + 4x(8) + 16 = x^4 + 8x^3 + 24x^2 + 32x + 16$.

9 [Intermediate] • Target: Binomial theorem

Find the coefficient of x^3 in the expansion of $(x+3)^5$, using $\binom{5}{k}$.

The term with x^3 comes from $\binom{5}{3}x^3(3)^2 = 10x^3(9) = 90x^3$. The coefficient is 90.

10 [Intermediate] • Target: Vectors in the plane

Given $\mathbf{u} = (3, -4)$, find its magnitude and a unit vector in the same direction.

$|\mathbf{u}| = \sqrt{9+16} = 5$. Unit vector: $\frac{1}{5}(3, -4) = (0.6, -0.8)$.

11 [Intermediate] • Target: A multi-part conic and function analysis problem

This question has two parts. A circle has equation $x^2 + y^2 - 4x + 6y - 12 = 0$. (a) Complete the square to write the equation in standard form, and state the center and radius. (b) Determine whether the point $(6, -1)$ lies inside, on, or outside the circle.

(a) Group and complete the square: $(x^2 - 4x + 4) + (y^2 + 6y + 9) = 12 + 4 + 9$, giving $(x-2)^2 + (y+3)^2 = 25$. Center: $(2, -3)$. Radius: 5. (b) Distance from center to $(6, -1)$: $\sqrt{(6-2)^2 + (-1-(-3))^2} = \sqrt{16+4} = \sqrt{20} \approx 4.47$. Since $4.47 < 5$, the point lies INSIDE the circle.

12 [Intermediate] • Target: Matrix operations

Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find $A - B$.

Subtract corresponding entries: first row $(2-0, -1-5) = (2, -6)$; second row $(3-(-2), 4-1) = (5, 3)$.

13 [Intermediate] • Target: Infinite geometric series

Find the sum of the infinite geometric series $8 + 4 + 2 + 1 + \cdots$, using $S = \frac{a_1}{1-r}$.

First term $a_1 = 8$, ratio $r = \frac{1}{2}$. Sum $= \frac{8}{1-\frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16$.

14 [Intermediate] • Target: Infinite geometric series

Write the repeating decimal $0.\overline{6}$ as a fraction, by treating it as an infinite geometric series $\frac{6}{10} + \frac{6}{100} + \frac{6}{1000} + \cdots$.

First term $a_1 = \frac{6}{10}$, ratio $r = \frac{1}{10}$. Sum $= \frac{6/10}{1-1/10} = \frac{6/10}{9/10} = \frac{6}{9} = \frac{2}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Piecewise and step functions in applied contexts • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A shipping company charges 5 dollars per pound for the first 10 pounds, then 3 dollars per pound for each pound beyond 10. Write a piecewise function $C(w)$ for the cost, and find the cost of a 15-pound package.

$C(w) = 5w$ for $0 < w \le 10$, and $C(w) = 50 + 3(w - 10)$ for $w > 10$. For 15 pounds: $C(15) = 50 + 3(5) = 50 + 15 = 65$ dollars.

16 [Challenging] • Target: Finding terms of a sequence defined recursively • Cross-Lesson: 7.01 Advanced Plane Geometry

A sequence is defined by $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \ge 2$. Find a_4.

$a_2 = 2(3) + 1 = 7$. $a_3 = 2(7) + 1 = 15$. $a_4 = 2(15) + 1 = 31$.

17 [Challenging] • Target: A multi-part conic section identification and analysis problem • Cross-Lesson: 8.01 Statistics, Counting & Probability

This question has three parts. An equation is given: $4x^2 + 9y^2 - 16x + 18y - 11 = 0$. (a) Complete the square in both x and y to write the equation in standard conic form. (b) Identify the type of conic section. (c) State the center and the lengths of the semi-axes.

(a) Group terms: $4(x^2 - 4x) + 9(y^2 + 2y) = 11$. Complete the square: $4(x^2 - 4x + 4) + 9(y^2 + 2y + 1) = 11 + 16 + 9$, giving $4(x - 2)^2 + 9(y + 1)^2 = 36$. Divide by 36: $\frac{(x-2)^2}{9} + \frac{(y+1)^2}{4} = 1$. (b) Since both terms are positive and added (not subtracted) with different denominators, this is an ellipse. (c) Center: $(2, -1)$. Semi-axes: $a = \sqrt{9} = 3$ (horizontal), $b = \sqrt{4} = 2$ (vertical).

18 [Challenging] • Target: Conic sections: hyperbolas • Cross-Lesson: 1.01 Advanced Number Theory

Find the 4th term in the binomial expansion of $(2x - 3)^6$.

Using the binomial theorem general term formula $T_{k+1} = \binom{n}{k} a^{n-k} b^k$ with $n = 6$, $a = 2x$, $b = -3$, and $k = 3$ (for the 4th term): $T_4 = \binom{6}{3}(2x)^{6-3}(-3)^3$. Compute the binomial coefficient: $\binom{6}{3} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$. $T_4 = 20 \cdot (2x)^3 \cdot (-27) = 20 \cdot (8x^3) \cdot (-27) = \mathbf{-4320x^3}$.

19 [Challenging] • Target: Parametric equations • Cross-Lesson: 2.01 Advanced Algebra & Systems

Given vectors $\vec{u} = \langle 3, -4, 2 \rangle$ and $\vec{v} = \langle 2, 1, k \rangle$, find the value of k such that $\vec{u}$ and $\vec{v}$ are orthogonal.

Two non-zero vectors are orthogonal if and only if their dot product equals zero: $\vec{u} \cdot \vec{v} = 0$ $(3)(2) + (-4)(1) + (2)(k) = 0$ $6 - 4 + 2k = 0 \Rightarrow 2 + 2k = 0 \Rightarrow 2k = -2 \Rightarrow \mathbf{k = -1}$

20 [Challenging] • Target: Sequences and series: sigma notation • Cross-Lesson: 3.01 Advanced Functions

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35$. Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256$. $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits: an introductory numerical approach

For the ellipse $\frac{x^2}{1681} + \frac{y^2}{81} = 1$ (horizontal major axis), find the coordinates of the foci.

a. $(\pm 9, 0)$

b. $(0, \pm 40)$

c. $(\pm 40, 0)$ ✓

d. $(\pm 41, 0)$

Since $a^2 = 1681$ and $b^2 = 81$: $c^2 = a^2 - b^2 = 1681 - 81 = 1600$, so $c = \sqrt{1600} = 40$. Foci: $(\pm 40, 0)$.

22 [Foundational] • Target: Average rate of change

A hyperbola has vertices at $(\pm 20, 0)$ and $b = 21$. Find the coordinates of the foci.

a. $(\pm 29, 0)$ ✓

b. $(\pm 20, 0)$

c. $(\pm 21, 0)$

d. $(\pm 41, 0)$

$c = \sqrt{a^2 + b^2} = \sqrt{400 + 441} = \sqrt{841} = 29$. Foci: $(\pm 29, 0)$.

23 [Foundational] • Target: Conic sections: ellipses

Find the sum of the infinite geometric series with first term $a_1 = 12$ and common ratio $r = \frac{1}{3}$, using $S = \frac{a_1}{1 - r}$.

a. 9

b. 36

c. 18 ✓

d. 8

$S = \frac{a_1}{1 - r} = \frac{12}{2/3} = 18$.

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24 [Foundational] • Target: Conic sections: hyperbolas

In the standard (x, y) coordinate plane, what is the equation of the horizontal asymptote of the graph of the function $f(x) = \frac{8x+3}{2x-5}$?

a. $y = 4$ ✓

b. $y = \frac{5}{2}$

c. $x = \frac{5}{2}$

d. $y = -\frac{3}{5}$

To find the horizontal asymptote of a rational function $f(x) = \frac{P(x)}{Q(x)}$ where the degrees of the numerator and denominator are equal, divide the leading coefficient of the numerator by the leading coefficient of the denominator: $y = \lim_{x \to \pm\infty} \frac{8x+3}{2x-5} = \lim_{x \to \pm\infty} \frac{8+\frac{3}{x}}{2-\frac{5}{x}} = \frac{8}{2} = 4$ Therefore, the horizontal asymptote is the line $y = 4$. Misconception checks: - $x = \frac{5}{2}$ is the vertical asymptote, foun.

25 [Foundational] • Target: Parametric equations

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n-1}{r-1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n-1}{2-1}$ $315 = 5\frac{2^n-1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

26 [Foundational] • Target: Sequences and series: sigma notation

What is the maximum value attained by the function $f(x) = \frac{\sin x - 2\cos x + 4}{2\sin x + \cos x + 3}$ for all real numbers x?

a. 3

b. 4

c. $\frac{9}{2}$

d. $\frac{11}{2}$ ✓

To find the maximum value of the rational trigonometric function, let $y = f(x)$: $y = \frac{\sin x - 2\cos x + 4}{2\sin x + \cos x + 3}$ First, check whether the denominator can ever equal zero. Using the linear combination identity for sine and cosine: $|2\sin x + \cos x| \le \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$ Since $\sqrt{5} < 3$, the denominator satisfies $3 - \sqrt{5} \le 2\sin x + \cos x + 3 \le 3 + \sqrt{5}$, meaning it is strictly positive for all real x. Multiplying both sides by the denominat.

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27 [Foundational] • Target: Binomial theorem

In the standard (x, y) coordinate plane, what is the equation of the vertical asymptote of the graph of $f(x) = \frac{3x+1}{x-4}$?

a. $y = 3$

b. $x = 4$ ✓

c. $x = -\frac{1}{3}$

d. $y = 4$

To find the vertical asymptote of a rational function in simplest form, determine the value of x that makes the denominator equal to zero while the numerator is non-zero. 1. Set the denominator equal to zero: $x - 4 = 0 \Rightarrow x = 4$ 2. Evaluate the numerator at $x = 4$: $3(4) + 1 = 13eq0$ Because the denominator is zero and the numerator is non-zero at $x = 4$, the line $x = 4$ is a vertical asymptote. Common Misconceptions: - $y = 3$ is the horizontal asymptote, found by taking the ratio of the leading c.

28 [Foundational] • Target: Vectors in the plane

What is the period of the function $f(x) = 3\cos(2x)$?

a. $\frac{\pi}{2}$

b. π ✓

c. 2π

d. 4π

The standard form of a cosine function is $f(x) = A\cos(Bx)$, where the amplitude is $|A|$ and the period is $\frac{2\pi}{|B|}$. For the function $f(x) = 3\cos(2x)$, the coefficient of x is $B = 2$. Applying the period formula: Period $= \frac{2\pi}{2} = \pi$ Common misconceptions: - Choice A ($\frac{\pi}{2}$) comes from incorrectly dividing π by B (the formula for the period of tangent). - Choice C (2π) forgets to account for the horizontal compression factor $B = 2$. - Choice D (4π) mista.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A multi-part conic and function analysis problem

A geometric sequence has first term $a_1 = 3$ and common ratio $r = 3$. Find a_4.

a. 9

b. -81

c. 243

d. 81 ✓

$a_4 = a_1 r^{4-1} = 3(3)^3 = 81$

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30 [Intermediate] • Target: Matrix operations

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.

a. **100** ✓

b. 110

c. 10

d. 210

Proof by Induction: Base Case ($n = 1$): $a_1 = 1$. The formula gives $1^2 = 1$. Thus, the base case holds. Inductive Hypothesis: Assume $a_k = k^2$ for some positive integer k. Inductive Step: We need to show that $a_{k+1} = (k+1)^2$. From the recurrence relation, $a_{k+1} = a_k + 2k + 1$. Using the inductive hypothesis, $a_{k+1} = k^2 + 2k + 1 = (k+1)^2$. Thus, by induction, $a_n = n^2$ for all positive integers n. Closed-form expression for S_N: $S_N = \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} n^2$. This is the su.

31 [Intermediate] • Target: Infinite geometric series

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. **2** ✓

b. 0

c. -2

d. 1

Let's compute the first few terms of the sequence: $a_1 = 2$ $a_2 = a_1^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ $a_3 = a_2^2 - 2 = 2^2 - 2 = 4 - 2 = 2$ It appears that for $a_1 = 2$, all subsequent terms of the sequence are also 2. We can prove this by induction. Base case: For $n = 1$, $a_1 = 2$, which is true. Inductive step: Assume that for some integer $k \geq 1$, $a_k = 2$. We need to show that $a_{k+1} = 2$. From the recurrence relation, $a_{k+1} = a_k^2 - 2$. By the inductive hypothesis, $a_k = 2$. Substituting this into.

32 [Intermediate] • Target: Piecewise and step functions in applied contexts

Find S_∞ for the geometric series with first term -8 and common ratio $\frac{1}{4}$.

a. $\frac{32}{3}$

b. $-\frac{32}{3}$ ✓

c. -6

d. $-\frac{32}{5}$

$S_\infty = \frac{a_1}{1-r} = \frac{-8}{1-\left(\frac{1}{4}\right)} = -\frac{32}{3}$

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33 [Intermediate] • Target: Finding terms of a sequence defined recursively

A geometric series has $a_1 = 3$ and $r = 2$. If $S_n = 21$, find n.

a. 5

b. 1

c. 4

d. 3 ✓

Testing successive partial sums against $S_n = a_1\frac{r^n - 1}{r - 1}$ gives $n = 3$.

34 [Intermediate] • Target: A multi-part conic section identification and analysis problem

A function is defined by $f(x) = 2x - 4$ for all x near $x = -4$ (including at $x = -4$). What is $\lim_{x \to -4} f(x)$?

a. -15

b. -12 ✓

c. -14

d. -10

f is a single continuous linear rule here, so the limit equals $f(-4) = -12$.

35 [Intermediate] • Target: Limits: an introductory numerical approach

The sequence $-7, -10, -13, -16, \ldots$ continues with the same pattern. Find the 14th term.

a. -43

b. -46 ✓

c. 46

d. -50

Arithmetic with $d = -3$: $a_{14} = -7 + (14 - 1)(-3) = -46$

36 [Intermediate] • Target: Average rate of change

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. **64** ✓

b. 96

c. 144

d. 192

Let $a_1 = b_1 = a$, let d be the common difference of (a_n), and let r be the common ratio of (b_n). 1. Express the given terms: $a_5 = a + 4d = b_2 = ar$ $a_6 = a + 5d = b_3 = ar^2$ Subtracting the first equation from the second gives: $d = a_6 - a_5 = ar^2 - ar = ar(r - 1)$ Substitute d back into $a + 4d = ar$:
$a + 4ar(r - 1) = ar$ Since the terms of the sequence are non-zero, $aeq0$. Dividing by a:
$1 + 4r(r - 1) = r \Rightarrow 4r^2 - 5r + 1 = 0$ $(4r - 1)(r - 1) = 0$ Since $b_2 eq b_1$.

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37 [Intermediate] • Target: Conic sections: ellipses

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

a. $\frac{256}{13}$ ✓

b. $\frac{128}{13}$

c. $\frac{64}{7}$

d. 16

For an infinite geometric sequence with first term $a_1 > 0$ and ratio $r \in (0, 1)$, the terms are $a_n = a_1 r^{n-1}$. 1. The sum of the original sequence is: $S_1 = \frac{a_1}{1-r} = 4 \Rightarrow a_1 = 4(1-r)$ 2. The terms of the squared sequence are $a_n^2 = a_1^2 (r^2)^{n-1}$, which forms a geometric sequence with first term a_1^2 and common ratio r^2. Its sum is: $S_2 = \frac{a_1^2}{1-r^2} = 8$ 3. Relate S_2 and S_1 using the algebraic identity $1 - r^2 = (1-r)(1+r)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Conic sections: hyperbolas • Cross-Lesson: 1.01 Advanced Number Theory

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$ ✓

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

Let the geometric sequence be denoted by a_n, where $a_1 = a$ and the common ratio is r. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $req1$). We are given that $S_3 = 21$ and $S_6 = 189$. So, we have the equations: 1) $\frac{a(r^3 - 1)}{r - 1} = 21$ 2) $\frac{a(r^6 - 1)}{r - 1} = 189$ We can rewrite $r^6 - 1$ using the difference of squares factorization: $r^6 - 1 = (r^3)^2 - 1^2 = (r^3 - 1)(r^3 + 1)$. Substitute this into the second equation.

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39 [Challenging] • Target: Parametric equations • Cross-Lesson: 2.01 Advanced Algebra & Systems

The first three terms of a geometric sequence of positive real numbers are $k-1$, $k+2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

b. 48

c. **96** ✓

d. 192

Step 1: Set up the ratio property for consecutive terms of a geometric sequence. Since the sequence is geometric, the common ratio r satisfies: $r = \frac{a_2}{a_1} = \frac{a_3}{a_2} \Rightarrow \frac{k+2}{k-1} = \frac{3k}{k+2}$ Step 2: Solve the quadratic equation for k. Cross-multiplying gives: $(k+2)^2 = 3k(k-1)$ $k^2 + 4k + 4 = 3k^2 - 3k$ $2k^2 - 7k - 4 = 0$ Factoring the quadratic: $(2k+1)(k-4) = 0$ Thus, $k = -\frac{1}{2}$ or $k = 4$. Step 3: Discard the extraneous solution. The problem stat.

40 [Challenging] • Target: Sequences and series: sigma notation • Cross-Lesson: 3.01 Advanced Functions

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155

b. 160

c. **315** ✓

d. 630

Step 1: Express the terms in terms of the first term a_1 and common ratio $r = 2$. The n-th term of a geometric sequence is given by $a_n = a_1 r^{n-1}$. For the second term: $a_2 = a_1(2)^{2-1} = 2a_1 = k \Rightarrow a_1 = \frac{k}{2}$ For the fifth term: $a_5 = a_1(2)^{5-1} = 16a_1$ Substituting $a_1 = \frac{k}{2}$ gives: $a_5 = 16\left(\frac{k}{2}\right) = 8k$ Step 2: Solve for the parameter k. We are given $a_5 = k^2 - 20$, so: $k^2 - 20 = 8k \Rightarrow k^2 - 8k - 20 = 0$ Factoring the quadratic equation.

41 [Challenging] • Target: Binomial theorem • Cross-Lesson: 4.01 Advanced Trigonometry

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n-1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n-1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n-1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n-1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

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42 [Challenging] • Target: Vectors in the plane • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. **3** ✓

c. 7

d. 9

Let the geometric sequence be $a_1, a_1r, a_1r^2, \ldots$. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a_1(1-r^n)}{1-r}$ (for $req1$). We are given that the sum of the first 3 terms is 21: $S_3 = \frac{a_1(1-r^3)}{1-r} = 21$ (Equation 1) We are also given that the sum of the first 6 terms is 189: $S_6 = \frac{a_1(1-r^6)}{1-r} = 189$ (Equation 2) We can rewrite Equation 2 using the difference of squares factorization, $1 - r^6 = (1 - r^3)(1 + r^3)$.

43 [Challenging] • Target: A multi-part conic and function analysis problem • Cross-Lesson: 7.01 Advanced Plane Geometry

In the standard (x, y) coordinate plane, the graph of $y = f(x)$ contains the point $(-2, 5)$. The function g is defined by $g(x) = f(x - 3) - 4$. Which of the following points must lie on the graph of $y = g(x)$?

a. $(1, 1)$ ✓

b. $(-5, 1)$

c. $(1, 9)$

d. $(-5, 9)$

We are given that the point $(-2, 5)$ lies on the graph of $y = f(x)$, which means $f(-2) = 5$. The function g is defined as $g(x) = f(x - 3) - 4$. To evaluate $g(x)$ using our known value $f(-2)$, we set the argument of f equal to -2: $x - 3 = -2 \Rightarrow x = 1$. Now, substitute $x = 1$ into the expression for $g(x)$: $g(1) = f(1 - 3) - 4 = f(-2) - 4 = 5 - 4 = 1$. Therefore, the point $(1, 1)$ must lie on the graph of $y = g(x)$. Geometrically, the transformation $y = f(x - 3) - 4$ translates the graph of.

44 [Challenging] • Target: Matrix operations • Cross-Lesson: 8.01 Statistics, Counting & Probability

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

c. 96 ✓

d. 192

Let a_1 be the first term and r be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1r + a_1r^2 + a_1r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for S_4: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for a_1.

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45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Advanced Number Theory

Consider an arithmetic sequence where the 5th term is 19 and the 11th term is 43. If the first term is denoted by a_1 and the common difference by d, find the value of $a_1 + d$.

a. 6

b. **7** ✓

c. 8

d. 9

Let the arithmetic sequence be defined by $a_n = a_1 + (n-1)d$. We are given that $a_5 = 19$ and $a_{11} = 43$. From the given information, we can set up a system of two linear equations: 1. $a_1 + (5-1)d = 19 \Rightarrow a_1 + 4d = 19$ 2. $a_1 + (11-1)d = 43 \Rightarrow a_1 + 10d = 43$ Subtract equation (1) from equation (2): $(a_1 + 10d) - (a_1 + 4d) = 43 - 19$ $6d = 24$ $d = \frac{24}{6} = 4$ Now substitute the value of d back into equation (1): $a_1 + 4(4) = 19$ $a_1 + 16 = 19$ $a_1 = 19 - 16 = 3$ We are aske.

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